

Junhyun (Jun) Kim

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INTRODUCTION

My research focuses on robot manipulation and vision-language-action models, particularly learning generalizable robot policies from human demonstrations. Previous experience includes 3D perception, surgical robotics, and medical image restoration.

EDUCATION

- **Georgia Institute of Technology** August 2025 - Present
M.S. in Robotics
 - Advisor: Prof. Zsolt Kira
 - GPA: 4.00/4.00
- **Seoul National University** March 2019 - August 2025
B.S. in Electrical and Computer Engineering
 - Summa Cum Laude, GPA: 3.99/4.30

SKILLS

- **Programming Languages:** C++, Python, Shell Script
- **Framework:** PyTorch, Open3D, ROS 2, LeRobot, MuJoCo
- **Tools:** MATLAB, Docker

WORK EXPERIENCE

- **Computer Vision Intern** January 2025 - June 2025
Sequor Robotics
 - Developed and optimized a prelabeling pipeline using 3D point cloud panoptic segmentation models and classical machine learning techniques, reducing manual labeling time by 75%.
 - Developed manual 3D labeling software for human annotators based on Open3D library

RESEARCH EXPERIENCE

- **Graduate Researcher** December 2025 - Present
Robotics Perception and Learning Lab in Georgia Tech
 - Developing demo-conditioned **Vision-Language-Action(VLA) Model** for robot manipulation
- **Undergraduate Research Assistant** May 2024 - July 2024
Advanced Robotic Technologies for Surgery Laboratory(ARTS Lab) in UT Austin
 - Improved the accuracy of single optic fiber shape sensing, which can be used for navigating body cavities during robotic surgery and biopsy.
 - * Resolved two critical hardware issues that caused large reconstruction errors
 - * **Reduced tip positioning error from 10~20 mm to under 7 mm**
- **Undergraduate Research Assistant** June 2023 - August 2023
Laboratory for Imaging Science and Technology(LIST) in Seoul National University
 - Utilized a **score-based diffusion model** to denoise MRI images of the substantia nigra(SN) region in human brain. A clean MR image for SN region is needed for better diagnosis of Parkinson's disease.
 - * Finetuned reverse steps to denoise SN region

PROJECTS

- **Teleoperation-Efficient Vision-Language-Action(VLA) Model** Fall 2025
Vision-Language-Model Course Project
 - Conditioned **human demonstration videos** on VLA to reduce data requirements when finetuning VLA on downstream embodiment.
 - * Obtained **53% Success Rate gain** by conditioning human videos at inference time.
 - To generalize our methods, we trained the VLA model without using cumbersome explicit retargeting methods
- **LLM Compression : Enhancing AWQ** Fall 2024
Graduation Project
 - Improved AWQ(Activation-aware Weight Quantization) with extra scaling
 - * Obtained lower perplexity(PPL) for INT3-quantized OPT and Llama 2